INCH-POUND

MIL-I-27189D(USAF)
15 OCTOBER 1993
SUPERSEDING
MIL-I-27189C(USAF)
28 FEBRUARY 1979

MILITARY SPECIFICATION

INDICATOR, PRESSURE, OIL

This specification is approved for use by the Department of the Air Force and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hermetically sealed, remote indicating, voltage ratio type integrally lighted, oil pressure indicators.

1.2 Classification. Indicators shall be of the following types, as specified (see 6.2):

MS24576-1	Red lighting
MS24576-2	White lighting
MS24576-3	White lighting with multilayer coated antireflection coverglass

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Beneficial comments)recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB, OK 73145-3037 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620
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Federal

PPP-B-636	Boxes, Shipping, Fiberboard
PPP-C-1797	Cushioning Material, Resilient, Low Density, Unicellular, Polyproylene Foam

Military

MIL-B-121	Barries Material, Greaseproofed, Waterproofed, Flexible
MIL-P-116	Preservation, Methods Of
MIL-C-675	Coating Of Glass Optical Elements (Anti-reflection)
MIL-I-25435	Transmitter, Pressure, Aircraft, Variable Reluctance Type, General Specification For
MIL-I-25437	Indicators, Aircraft, Pressure, Voltage Ratio Type, General Specification For
MIL-I-25467	Lighting, Integral, Red, Aircraft Instrument, General Specification For
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification For
MIL-T-38230	Transmitter, Pressure TRU-66/A Oil, Variable Reluctance

STANDARDS

Federal

FED-STD-101	Preservation, Packaging And Packing Materials: Test Procedures
FED-STD-595	Color

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-785	Reliability Program For Systems And Equipment, Development And Production

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MIL-STD-2073-1	DOD Material Procedures For Development And Application Of Packaging Requirements
MS24576	Indicator, Pressure Oil
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Building #4, Section D, Philadelphia, PA 19111-5094.)

3. REQUIREMENTS

3.1 Qualification. The indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 General specification. The requirements of MIL-I-25437 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.2.2 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification in. I the event of any conflict between the requirements of this specification and the specification sheet, the later shall govern.

3.3 Design and Construction.

3.3.1 Indicator. The indicator shall be designed for use with a TRU-66/A transmitter conforming to MIL-T-38230.

3.3.2 Case. The case dimensions shall conform to MS33639 for the 2-inch size. The length of the case shall not exceed 2.75 inches.

3.3.2.1 Cover Glass.

3.3.2.1.1 Perfect Reflectance (MS24576-3). All reflecting glass surfaces of the MS24576-3 indicator shall be provided with a reflection reducing coating which meets the requirements of MIL-C-675 with the exception of the materials used, in addition to withstanding the environmental conditions specified herein. The following reflectance tolerances shall apply:

Angle of Incidence	Wavelength Millimicrons	Percent Reflectance
0°	450-675	0.6 absolute
0°	425-700	0.5 average
30°	400-625	1.0 absolute
30°	425-700	0.5 average

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3.3.3 Reliability program. The contractor shall establish a reliability program in accordance with MIL-STD-785.

3.3.3.1 Reliability. The indicator shall have a minimum acceptable mean-time-between-failures (MTBF) of 2,000 hours based upon a discrimination ratio of 1.5 and confidence factor of 90 percent.

3.3.3.2 Longevity. The indicator shall have an operating life span of 3,000 hours.

3.4 *Performance*.

3.4.1 Tolerances. The indicator shall operate within the tolerances specified in tables I and II.

3.5 Lighting. The indicators shall be integrally lighted and shall meet all requirements specified herein. The MS24576-1 indicator shall be lighted in accordance with MIL-L-25467 and the MS24576-2 and MS24576-3 indicators shall be lighted in accordance with MIL-L-27160.

3.6 Pointer. The pointer shall be in accordance with figure 1 except that the length shall be such that the tip will extend into the scale a distance equal to approximately one-third the length of the shortest graduation. The pointer shall be suitable for integral lighting.

3.6.1 Pointer finish. The lighted portion of the pointer shall be finished in accordance with FED-STD-595, color No. 37875.

3.7 Dial. The dial shall conform to figure 1 and shall be of material suitable for integral lighting. A minimum diameter of 1.75 inches shall be maintained across the outside of the graduations. The graduation for 45 psi shall be located on the horizontal centerline at the left side of the dial.

3.7.1 Markings.

3.7.1.1 White. The following markings shall be finished in durable white, color No. FED-STD-595. The dimensions shall be as specified.

	Height or Length Inch +0.016	Width of Line or Graduation Inch +0.005
Numerals 0, 20, 40, 60, 80, and 100	0.157	0.025
10-pound graduations	0.188	0.031
5-pound graduations	0.094	0.020
Lettering OIL PRESS	0.125	0.025
Lettering PSI	0.125	0.025

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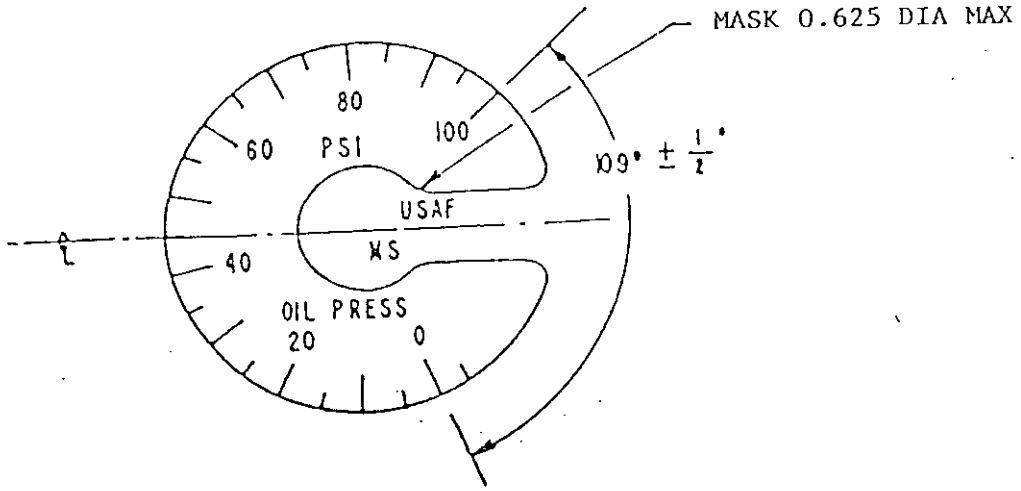
TABLE I. Temperature Scale Error.

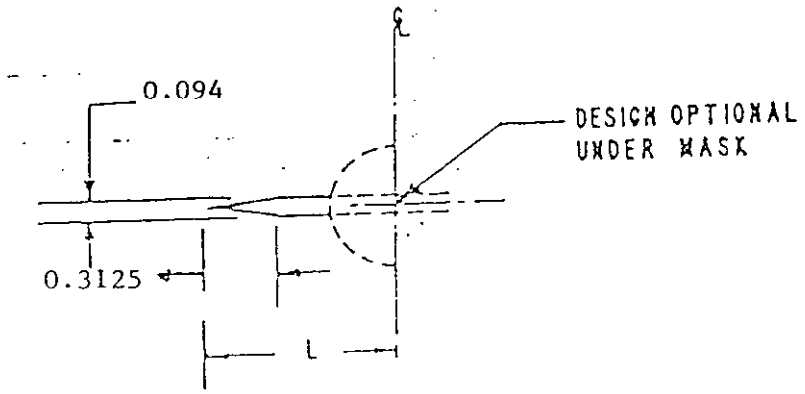
Test Pressure PSI	Room Temperature PSI	Scale Error High and Low Temp Exposure High and Low Temperature PSI
0	1.5	3.00
10	1.5	3.00
20	1.5	3.00
30	1.5	3.00
40	1.5	3.00
50	1.5	3.00
60	1.5	3.00
70	1.5	3.00
80	1.5	3.00
90	1.5	3.00
100	1.5	3.00

TABLE II. Position, Friction, and Vibration Error.

Test	Tolerance +PSI
Position error at 50 psi	1.5
Friction	
Room temperature	1.5
High temperature	2.0
Low temperature	2.25
Vibration	
Pointer oscillation	2.0
Pointer variation	1.5

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DIMENSIONS IN INCHES

UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMAL 0.016

FIGURE 1. Oil Pressure Indicator.

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3.7.1.2 BLACK. The markings USAF and MS24576 (*) (applicable dash No.) shall be permanently and legibly marked on the mask in letters 1/16 inch in height. Unless otherwise specified, these and all other markings shall be finished in durable dull black, color No. 37038 of FED-STD-595.

3.8 Electrical connector. The indicator shall be provided with an electrical connector, Deutsch Company part No. DM5600-7P, or equal, to mate with Deutsch Company part No. DM9700-7S, or equal. The connector shall be furnished with a cap or cover and gasket to prevent penetration of water vapor.

3.9 Electrical wiring. The internal and external wiring shall conform to figure 2. The internal wiring shall be color coded and shall be connected to the receptalce pins as shown.

3.10 Electrical characteristics. The current drawn by the indicator shall not exceed 100 ma with pins 3, 6, and 7 open circuited. The power consumption of the indicator, exclusive of lamps, shall not exceed 1w.

3.11 Weight. The weight of the indicator shall not exceed 0.50 pound.

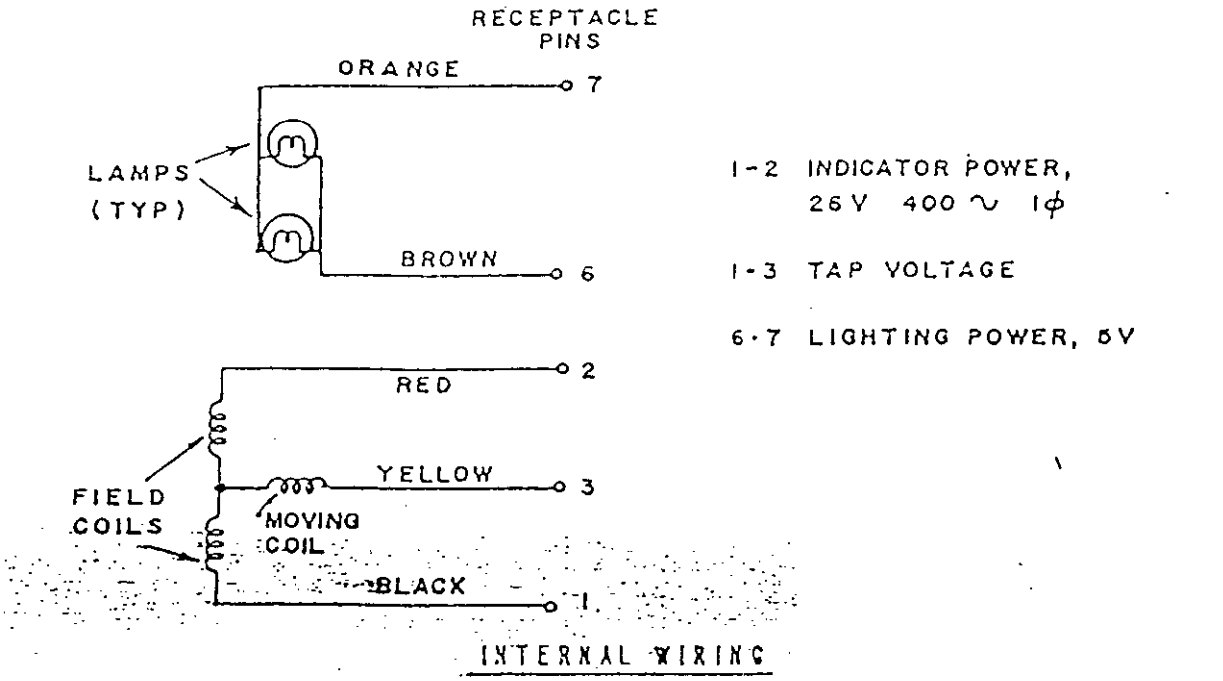
3.12 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and test) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of know defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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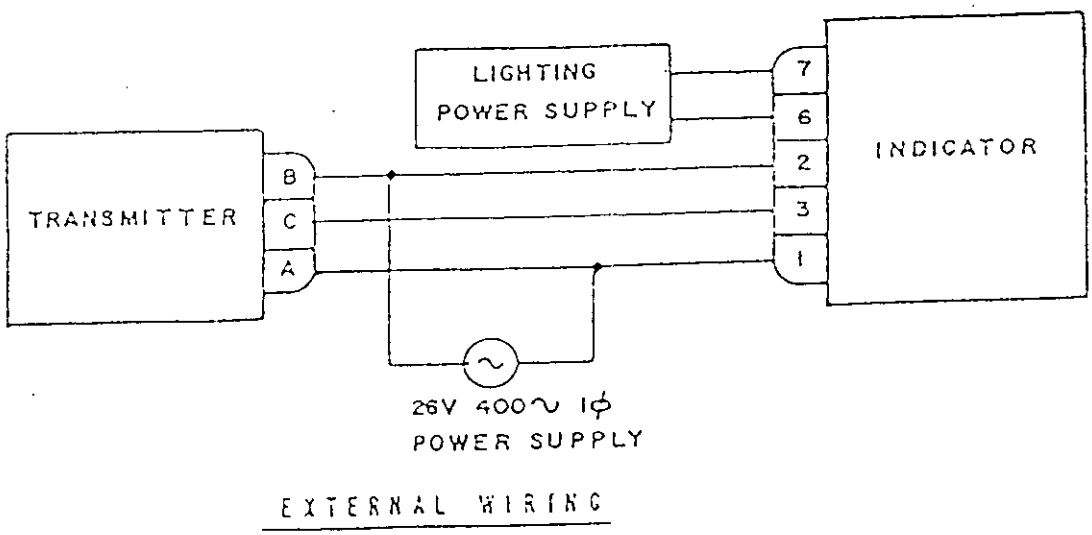


FIGURE 2. Wiring diagram.

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4.2 Classification of inspections. The inspection requirements specific herein are classified as follow:

a. Qualification inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.2.1 General specification. The inspection and testing of indicators shall be in accordance with MIL-I-25437 and as specified herein. The indicators shall operate within the tolerances specified in 3.4.1.

4.3 Qualification inspection.

4.3.1 Test samples. The qualification test samples shall consist of ten indicators representative of the production equipment. Three of the samples shall be subjected to the reliability test only. Samples shall be identified manufactures own part number and any additional information required by the letter of authorization.

4.3.2 Qualification tests. The Qualification tests shall consist of all the tests specified herein and in MIL-I-25437.

4.4 Test methods.

4.4.1 Lighting. The MS24576-1 indicator shall be subjected to lighting tests in accordance with MIL-L-25467 and the MS24576-3 indicators shall be subjected to lighting tests in accordance with MIL-L-27160.

4.4.2 Reliability.

4.4.2.1 Demonstration. Unless otherwise specified, seven indicators shall be subjected to the reliability test specified in 4.4.2.2 to demonstrate a minimum required MTBF of 2,000 hours at a confidence level of 90 percent. The total test time, number of failures permitted, and accept, continue-test, reject criteria shall be in accordance with figure 3.

4.4.2.2 Test. The test equipment and fixtures shall include provisions for vibration and temperature cycling at the periodic intervals specified herein. During this test, input signals shall be applied to the indicator at a rate of 2 to 3 cycles (minimum and maximum) per minute from 0 scale to 85 psi and return. The test shall begin when rated power on, which shall include at least 480 cycles of operation, the power shall be turned off and shall remain off for a period of 10 minutes. For at least 10 minutes out of every hour of ON time, the indicator shall be vibrated in accordance with MIL-STD-781, test level J. The temperature cycles shall be conducted in the following sequence: 4 hours at -20° C, 14 minutes for temperature change to +50° C at a minimum rate of 5° C per minute; 4 hours at +50° C, 14 minutes for temperature change to -20° C; and 4 hours at -20° C. In case of malfunction, the malfunction shall be recorded and the test enterrupted while repairs are made. Upon completion of each 250 hours of cycling, the indicator shall be

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checked for scale error. The scale error reading shall not exceed 150 percent of the room temperature tolerance in table I. No recalibration shall be performed on the units during this test. The test may be interrupted for weekend shutdown. The contractor shall prepare a report in accordance with MIL-STD-781 which shall contain test procedures, details of any malfunctions, and corrective actions taken.

4.4.2.3 Longevity. At the completion of the reliability test, three of the indicators shall continue to be tested for a total of 3,000 hours each. The test time accumulated during the reliability test shall be applied to this test. The accept-reject criteria is not applicable.

4.4.3 Compatibility. One indicator shall be submitted to the preparing activity for a compatibility test. The indicator shall be connected to a variety of transmitters which meet the requirements of MIL-T-38230. The indicator and transmitter shall be connected per Figure 3 of MIL-T-25435. The scale error test of MIL-T-25435 shall be performed at room temperature. The tolerance at each test point shall be the sum of the scale error tolerances of MIL-T-38230 and MIL-I-27189 at room temperature.

4.5 Inspection and test. Tests of methods of preservation shall be accomplished in accordance with Section 4 of MIL-P-116, to insure compliance with Section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C" as specified (see 6.2).

5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in Quantity Unit Pack (QUP) of one each and Intermediate Container Quantity (ICQ) of twelve (12) each. Each item will be provided a preservation method IC-1 In Accordance With (IAW) MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned IAW MIL-P-116 C-1 process.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved IAW MIL-P-116 and MIL-STD-2073-1 to provide a method IC-1.

5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073-1.

5.2 Packing. Packing shall be level "A", "B", or "C", as specified (see 6.2).

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5.2.1 Container-Cushioning. Unit container requirements shall conform to MIL-B-121, barrier material, unless otherwise specified by the contracting activity. Unit container shall be large enough to allow for application of sufficient cushioning/wrap material, between the item and unit container. Item cushioning/wrap material shall conform to PPP-C-1797 and provide sufficient density and thickness to protect the unit container from punctures. Intermediate containers requirements shall conform to PPP-B-636, weather-resistant unless otherwise specified by the contracting activity. Intermediate containers be large enough to accommodate twelve (12) each unit containers. Cushioning material shall be of sufficient density and thickness to insure shock transmission does not exceed peak values in G's established for the item.

5.2.2 Level A. Fiberboard containers do not meet Level "A" container criteria.

5.2.3 LEVEL B. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-636, weather-resistant, unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size and minimum tare and cube, consistent with the protection required. Closure shall be IAW appropriate PPP-B-636 procedures, as specified by contractor, Special Packaging Instructions (SPI), and contracting activity.

5.2.4 Level C. Item will be packed in such a manner that will afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-2073-1.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked IAW special markings required by the contractor, SPI and MIL-STD-129.

5.4 Inspection and Test. Test of methods of preservation shall be accomplished IAW section 4 of MIL-P-116 to insure compliance with section 5 of this specification. Packaging tests shall be conducted IAW rough handling as specified in FED-STD-101.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The indicators covered by this specification are intended for use inconjunction with the TRU-66/A pressure transmitter in indicating remotely the pressure in oil systems of aircraft engines.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

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b. Type of indicator required (see 1.2)

c. Levels of packaging and packing required (see section 5)

d. Issue of Department of Defense Index of Specifications and Standards (DODISS) to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1)

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City Air Logistics Center/TICLA, Tinker AFB, OK 73145-3037 and any information pertaining to qualification of products may be obtained from this activity.

6.4 Subject term (key word) listing.

Lighted
MIL-T-38230
Variable reluctance
Voltage ratio
0-100 PSI

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
AIR FORCE - 99

Preparing Activity:
AIR FORCE - 71

Project Number:
6620-F508

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I27189D	93/10/15

3. DOCUMENT TITLE
INDICATOR, PRESSURE, OIL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME QC-ALC/TICLA	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code) TINKER AFB OK 73245-3037	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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